Enhancing Legalyze: Automated Data Synthesis and Structure-Aware Retrieval-Augmented Generation for Pakistani Legal Intelligence

A Research-Direction and Results-Design Template

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Abstract

This template sets out a proposed research direction that extends the Legalyze platform (Hamza et al., 2026) by transferring the automated data-synthesis and fine-tuning methodology of Ren et al. (2026) from Chinese to Pakistani law, and fusing it with Legalyze's structure-aware hierarchical indexing. The central hypothesis is that a well-adapted generator operating over a structure-aware index outperforms either a purely model- and data-centric pipeline or a purely structure-centric one. The document states enhanced research questions, maps each component of the base methodology onto the Pakistani setting, and specifies a results-focused experimental design, including a baseline ladder, separated retrieval and answer-quality metrics, planned ablations, and empty result tables (T1–T5) to be completed with measured values. All cited records are drawn from the project bibliography; no results are reported here, as every number in the tables is a placeholder.

1 Motivation and positioning

The base paper for this direction is Ren et al. (2026), *Automated Data Synthesis and Retrieval-Augmented Generation for Legal Large Language Models*. Its pipeline is model- and data-centric: it (i) generates synthetic legal question–answer data, (ii) curates that data with an LLM-based judge and an enhanced Elo-ranking mechanism, (iii) fine-tunes the retriever and generator on the curated data, and (iv) evaluates the result on a Chinese legal benchmark (LawBench).

Legalyze (Hamza et al., 2026) follows the opposite, structure-centric philosophy for Pakistani law: structure-aware hierarchical indexing (SAHI), a hybrid of BM25 and dense retrieval fused by reciprocal rank fusion (Cormack et al., 2009), and an agentic, citation-grounded response design.

The proposed direction is therefore not a simple re-run of Ren et al. (2026) on Pakistani data. Its contribution is the *combination*: transferring the data-synthesis and fine-tuning methodology to a low-resource jurisdiction and integrating it with Legalyze's structure-aware retrieval, then testing whether the combination improves evidence-grounded answer quality over either philosophy alone. This mirrors the synthesis argued for in the accompanying survey, where model-centric designs (Buffa et al., 2025; Ren et al., 2026) and the structure-centric design (Hamza et al., 2026) are presented as complementary rather than competing.

2 Enhanced research questions

Table 1 states the research questions that extend Legalyze with the base methodology. Each is designed to be answered by a controlled experiment in Section 4.

3 Methodology transfer

Table 2 maps each component of the base methodology onto the Pakistani-Legalyze setting, with the principal risk to control at each step. The guiding principle is to keep legal evidence external, in the SAHI index, so that fine-tuning adapts *phrasing and retrieval* rather than internalising legal knowledge in model weights, which would reintroduce the hallucination risk documented by Dahl et al. (2024) and Magesh et al. (2025).

4 Experimental and results design

This section is the focus of the proposed work. It separates retrieval quality from answer reliability, reports results only within their own experimental context, and uses ablations to attribute any gain to a specific component. These disciplines follow the evaluation guidance

Table 1: Enhanced research questions for Legalyze under the Ren et al. (2026) methodology.

RQ	Focus	What the experiment must establish
RQ-A	Automated synthetic legal QA generation for a low-resource jurisdiction	Whether an LLM, grounded in SAHI-retrieved Pakistani provisions and judgments, can produce legally faithful QA pairs usable for fine-tuning.
RQ-B	Data curation (LLM-judge + Elo ranking)	Whether curating synthetic data improves downstream retrieval and answer quality relative to unfiltered synthetic data.
RQ-C	What to fine-tune	Whether fine-tuning the retriever only, the generator only, or both yields the best citation precision and legal correctness on Pakistani law.
RQ-D	The fusion hypothesis (headline)	Whether SAHI combined with synthetic-data fine-tuning outperforms SAHI alone and a Ren et al. (2026)-style fine-tuned pipeline over flat chunks.
RQ-E	Multilingual access	How gains differ across English, Urdu, and Roman Urdu queries.

Table 2: Component-by-component transfer of the Ren et al. (2026) methodology to Pakistani Legalyze.

Base component	Pakistani-Legalyze adaptation	Principal risk and control
Synthetic QA generation	Prompt an LLM with SAHI-retrieved provisions (Constitution, PPC, CrPC, QSO) and judgments; generate question, answer, and the breadcrumb as the gold citation.	LLM hallucinating Pakistani law; control by grounding every synthetic answer in a retrieved passage and rejecting ungrounded items.
LLM-judge + Elo curation	Judge on legal correctness and citation support, not fluency; auto-verify the cited provision against the SAHI ground truth; Elo-rank candidates.	Judge and generator sharing a model family; control by using a different judge model plus a deterministic citation-resolution check.
Fine-tune retriever	Parameter-efficient (for example LoRA) contrastive fine-tuning of the dense embedder on synthetic (query, gold-passage) pairs; retain BM25 and RRF.	Compute cost; control with parameter-efficient fine-tuning and a retriever-first strategy.
Fine-tune generator	Light parameter-efficient adaptation only; keep evidence external via SAHI.	Re-internalising knowledge in weights; control by measuring hallucination before and after fine-tuning.
Benchmark (LawBench analogue)	Construct a held-out, expert-annotated Pakistani legal QA set (PakLawBench) with statute-retrieval, citation, and application tasks, split by language.	No public analogue exists; the benchmark itself is a contribution and must document coverage and annotation.

synthesised in the survey and the evidence that semantic similarity does not establish legal correctness (Silva and Sanfilippo, 2026), and that failures arise from both retrieval and reasoning (Afane et al., 2026).

4.1 Baseline ladder

Table 3 defines the systems to be compared. Each step up the ladder isolates one idea, so that differences between adjacent rows are interpretable.

4.2 Metrics

Metrics are reported in separate families and never collapsed into a single score.

- **Retrieval:** Recall@k, nDCG@k, MRR, and passage-level precision.
- **Answer quality:** expert-assessed legal correctness, citation precision, citation recall, citation-support (whether the cited passage entails the claim), and unsupported-claim or hallucination rate measured under a declared protocol (Dahl et al., 2024; Magesh et al., 2025).
- **Failure attribution:** the share of errors due to retrieval misses versus reasoning errors (Afane et al., 2026).
- **Multilingual:** every metric above, reported separately for English, Urdu, and Roman Urdu.
- **Operational:** latency and the compute or service cost of fine-tuning and inference.

Each reported number should carry its annotation protocol, reviewer expertise, inter-annotator agreement, and, where applicable, a significance test. Retrieval-benchmark design follows the precedent of passage-level legal retrieval evaluation (Pipitone and Houir Alami, 2024; Butler and Butler, 2026), and legal benchmark construction follows (Chalkidis et al., 2022; Guha et al., 2023).

4.3 Planned result tables

All cells are placeholders (–) to be completed with measured values; no results are reported in this template.

4.4 Hypotheses to be tested

The following are stated in advance and then tested; they are not reported findings.

Table 3: Baseline ladder. Each row adds one component relative to the row above.

ID	System	What it isolates
B1	Base LLM, no retrieval	Parametric-only baseline; lower bound on traceability.
B2	Vanilla RAG (flat chunks, off-the-shelf retriever)	Effect of retrieval alone, without structure or fine-tuning.
B3	Ren et al. (2026)-style: flat chunks + synthetic-data fine-tuned retriever and generator	The base methodology on Pakistani law, without structure.
B4	Legalyze SAHI (structure-aware, no fine-tuning) (Hamza et al., 2026)	Effect of structure-aware indexing alone.
B5	**Proposed:** SAHI + curated synthetic-data fine-tuning	The fusion of structure and the base methodology.

Table 4: **T1 – Retrieval quality** across baselines. Higher is better. R@k: Recall@k; P@5: passage precision at 5.

System	R@5	R@10	nDCG@10	MRR	P@5
B1 Base LLM (no retrieval)	–	–	–	–	–
B2 Vanilla RAG (flat)	–	–	–	–	–
B3 Ren-style fine-tuned (flat)	–	–	–	–	–
B4 Legalyze SAHI (no FT)	–	–	–	–	–
B5 **SAHI + synthetic FT (ours)**	–	–	–	–	–

1. B5 gives the largest gain on exact-statute citation tasks, because structure aids retrieval while fine-tuning aids phrasing.
2. B3 (model-centric fine-tuning over flat chunks) improves fluency but not citation precision, and may raise the hallucination rate, directly testing the survey's critique of model-centric designs.
3. Curation (LLM-judge and Elo) matters more in the low-resource Pakistani setting than in a data-rich setting.
4. Gains are smaller for Urdu and Roman Urdu unless terminology alignment is added, motivating the multilingual work.

4.5 Fully automatic evaluation without human annotation

A central design goal of this proposal is to evaluate the system *without* a human annotation team. This is possible for two reasons. First, because the corpus is structure-aware, the gold passage and gold citation for every synthetic item are known by construction from the breadcrumb that generated it, so retrieval and citation metrics need no human labels. Second, the base methodology (Ren et al., 2026) already evaluates with an LLM-based judge and an Elo-ranking mechanism rather than a human panel, and a recent legal-RAG study used an LLM-as-judge validated against only a small expert sample (Silva and Sanfilippo, 2026). Removing human annotation is therefore more faithful to the base method, not a compromise. Table 9 shows how each metric is obtained automatically.

A single optional human step preserves trust in the automatic judge: on a small sample of roughly fifty items, the LLM-judge and one reviewer score the same answers and the agreement is reported (for example Cohen's κ). This takes a few hours rather than the hundreds required for full annotation, and it can be omitted, in which case the evaluation is reported as fully automatic. The honest framing is that legal correctness is assessed automatically by a held-out judge, with large-scale expert evaluation left to future work; this is a disclosed limitation, not a flaw in the contribution.

All stages are intended to run on a local compute cluster, so no cloud GPU rental is required. Parameter-efficient fine-tuning (for example QLoRA on a 7–8B open model) and a low-cost API model used under the Batch API for synthetic generation and judging keep the monetary cost minimal.

5 Contribution and honest-reporting discipline

The contribution is threefold: (i) the first application of an automated data-synthesis and quality-curation methodology to a low-resource jurisdiction; (ii) the integration of that methodology with structure-aware hierarchical indexing, producing a structure-aware and data-synthesis hybrid; and (iii) a multilingual, results-focused evaluation that separates retrieval, citation, and legal correctness across English, Urdu, and Roman Urdu. Because fine-tuning can re-internalise knowledge, the hallucination rate is measured before and after fine-tuning and evidence is kept external in the SAHI index, so a known risk becomes a reported finding rather than a hidden assumption. Any comparison with the existing Pakistani RAG framework (Ali et al., 2025) requires a shared query set and corpus before a claim of superiority can be made.

Table 5: **T2 − Answer quality and reliability** across baselines. For hallucination rate, lower is better; all other columns, higher is better.

System	Legal corr.	Cit. prec.	Cit. recall	Cit. support	Halluc. rate
B1 Base LLM (no retrieval)	–	–	–	–	–
B2 Vanilla RAG (flat)	–	–	–	–	–
B3 Ren-style fine-tuned (flat)	–	–	–	–	–
B4 Legalyze SAHI (no FT)	–	–	–	–	–
B5 **SAHI + synthetic FT (ours)**	–	–	–	–	–

Table 6: **T3 − Ablation.** Each row removes or changes one component of B5 to attribute its effect.

Configuration	R@10	Cit. prec.	Legal corr.	Halluc. rate
B5 full (SAHI + curated synthetic FT)	–	–	–	–
fine-tune retriever only	–	–	–	–
fine-tune generator only	–	–	–	–
unfiltered synthetic data (no Elo curation)	–	–	–	–
flat chunks instead of SAHI	–	–	–	–
no RRF (single retrieval stream)	–	–	–	–

6 Risks and mitigations

- **Synthetic Pakistani law may be incorrect.** Generate only from retrieved passages, auto-verify the gold citation, and human-check a sample.
- **Fine-tuning cost.** Use parameter-efficient fine-tuning and prioritise the retriever, which is the cheapest change with the largest expected citation gain.
- **Fabricated evaluation numbers.** Every table here is a design; publish only measured results with agreement statistics and significance tests.
- **Unfair cross-study comparison.** Do not compare against results from other jurisdictions or protocols without re-running under a common setting.

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Table 7: **T4 – Multilingual breakdown** for the proposed system (B5), by query language.

Query language	R@10	Legal corr.	Cit. prec.
English	–	–	–
Urdu	–	–	–
Roman Urdu	–	–	–

Table 8: **T5 – Synthetic-data scaling** for the proposed system (B5): quality versus the number of curated synthetic training examples.

# examples	R@10	Legal corr.
1,000	–	–
5,000	–	–
10,000	–	–
50,000	–	–

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Table 9: Fully automatic sources of ground truth, replacing human annotation.

Metric family	**Automatic source of ground truth**
Retrieval (Recall@k, nDCG, MRR)	Gold passage is known from the generating provision; computed by code with no labels.
Citation existence and resolution	Deterministic lookup in the indexed corpus.
Citation precision and recall	Compare the system's cited passage with the known gold passage.
Citation support (entailment)	A natural-language-inference model or an LLM-as-judge.
Legal correctness	LLM-as-judge against the known gold answer, as in the base methodology (Ren et al., 2026).
Hallucination / unsupported-claim rate	A citation that does not resolve is fabricated; a claim not entailed by its citation is unsupported.
Multilingual consistency	Metamorphic test: the same question in English, Urdu, and Roman Urdu should retrieve the same provision.